VF-E19 Hydrolysis Test

Test Method for Hydrolysis Test	Test method # VF- E19 REF #:
Date Issued: December 2, 2024 Revision #: 5	Reason for Revision: Revise evaluation for Nubuck/Suede, metal parts

PURPOSE AND SCOPE: To determine the change of physical and mechanical properties as well as the physical appearance through artificial ageing.

REFERENCE: DIN 53543, SATRA TM 344, ASTM D3137

EQUIPMENT AND MATERIAL:

1. Air-circulating oven, capable of maintaining temperature of 60 ± 1°C and relative humidity at 95 ± 1%.
2. Tensile machine and other lab equipment required for hydrostatic stability test.
3. PU material, PU coated material or with prints, Rubber, TPU, Footwear.
4. Distilled or pure water with pH 6.5 ±1.0

SAMPLE PREPARATION:

Condition the test samples at 23 ± 2°C and 55 ± 10% relative humidity for at least 24 hours prior to testing. Samples with water base cement should be conditioned for 48 hours after assembly unless the cement supplier has specified that 24 hours is already sufficient for conditioning period.

REPETITIONS or QUANTITY of SAMPLE:

1. For normal hydrolysis test:
   - Footwear: 1 sample per material combination.
   - For material / component: 2 samples minimum
2. For hydrolytic stability:
   - Footwear: 1 pair per material combination.
   - For material / component: 6 samples minimum

TEST PROCEDURE:

A. For regular hydrolysis test:

1. Place the test material / footwear in an oven at 60 ± 1°C for 7 days at 95 ± 1 % relative humidity. ***Ensure to use* distilled or pure water *with pH 6.5 ±1.0 to avoid deposit on the surface of the tested sample that can be mistaken as blooming or corrosion.***
2. After 7 days, remove the test material / footwear from the oven.
3. After removing the specimen from the oven, let it stabilize for 24 hours at 23± 2°C and 55 ±10% relative humidity.
4. After conditioning the test sample:

- Check delamination of components or peel-off in case of ink
- Check change of appearance – blooming, color change, color migration, bleeding, bubbles, blisters, etc.

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B. For Hydrolytic Stability Test:

- **For Material / Component:**
  1. Cut at least 6 specimens for tensile test and elongation test. ***(Note: May also prepare other test specimens if other physical properties are required such as for bally flex)***
  2. On the 3 pieces conduct tensile and elongation test or other physical properties required. These data are the original properties. ***(Note: For bally flex, no need for original properties)***
  3. On the remaining 3 pieces for hydrolytic stability test follow **steps B :1-3**
  4. After conditioning, conduct the required physical properties
- **For footwear:**
  1. Follow **steps A: 1-3**
  2. After conditioning, conduct the required test.

REPORT:

1. **For regular hydrolysis test.**
   - Change of physical appearance. ***Refer to material specification on the requirement per type of material.***
   - Ink/Print: ***Material specification requirement - NO delamination*** **or peel off**
   - Adhesion: ***No delamination.*** **No need to evaluate delamination which cause by material shrinkage**
   - **Color change: Nubuck/Suede no evaluation**
   - **Color bleeding/color migration: No color bleeding / color migration**
   - **Metal corrosion /rust: no evaluation**
2. **Blooming Test:** No blooming
3. **For Hydrolytic Stability:**

-% Retention of physical properties. ***Refer to material specification on the requirement.***

Note: Hydrolytic Stability test is optional. This is only needed in case there's an issue on the quality of footwear or if required by the brand. ***But for TPU, PU strap and component this is mandatory***.